

Assignment 2 – Case Study Lab

Stage 2 Lab Activities

SmartCare Requirements Engineering

AI OFF -> AI ON -> VERIFY | 1 hour

Learning objectives

- Analyse the SmartCare client brief.
- Identify stakeholders and scope.
- Write functional and non-functional requirements.
- Develop user stories and Given-When-Then acceptance criteria.
- Use AI to critique requirements without allowing it to invent stakeholder needs.
- Produce SmartCare Requirements Specification v1.0.

Part A - Client Brief: AI OFF

SmartCare uses spreadsheets and paper records. Staff report duplicate bookings, difficulty finding patient information, inconsistent appointment status and limited appointment history. Management wants a small, maintainable patient, practitioner and appointment system.

Part B - Stakeholders and Scope: AI OFF

Identify at least four stakeholders. Create In Scope and Out of Scope lists. Label uncertain features as provisional rather than confirmed.

Stakeholders:

- Patients
- Practitioners
- Reception staff
- Clinic management
- System administrator

In Scope

Managing patient information

Managing practitioner information

Booking appointments

Cancelling appointments

Out of scope

Online payments - Provisional

Facial recognition - Provisional

AI treatment recommendations - Provisional

Complex hospital management - Provisional

Managing appointment status	Advanced clinical/medical functions
Preventing duplicate bookings	SMS/email appointment reminders - Provisional
Retaining cancelled appointments in appointment history - Provisional	Full hospital information system
Viewing practitioner availability	Features unrelated to patient, practitioner and appointment management - Provisional
Producing basic operational reports	
Searching for patient records by ID	

Part C - Functional Requirements: AI OFF

Write 8-12 numbered functional requirements using FR-01, FR-02 and so on. Each should describe one observable capability.

FR-01: The system shall allow staff to create a new patient record.

FR-02: The system shall allow staff to search for a patient by name.

FR-03: The system shall allow staff to update patient information.

FR-04: The system shall allow staff to create an appointment for a patient with a practitioner.

FR-05: The system shall prevent an appointment from being created when the selected practitioner is already booked for that time.

FR-06: The system shall allow staff to view a practitioner's scheduled appointments and availability.

FR-07: The system shall allow staff to cancel an appointment.

FR-08: The system shall allow staff to update the status of an appointment.

FR-09: The system shall generate basic operational reports from stored appointment data.

Part D - Non-Functional Requirements: AI OFF

Write 4-6 numbered non-functional requirements covering appropriate qualities such as reliability, maintainability, usability, data integrity or testability.

01: The system shall maintain appointment and patient data without crashing during operation.

02: The system will allow staff to complete simple task like searching a patient name without having the technical knowledge.

03: The system should be consistent with the patient, practitioner, staff and with appointment information so that an appointment cannot be associated with a non-existent patient or practitioner.

04: The system shall use a modular design so that individual components can be modified or maintained without requiring changes to unrelated components.

05: Core business logic should be independent to be tested without requiring the application to run.

Part E - User Stories and Acceptance Criteria: AI OFF

Write 4-6 user stories. For at least three, create Given-When-Then acceptance criteria including one negative or failure scenario.

Story 1

As a receptionist, I want to create a patient record so that patient information can be stored and retrieved by the clinic.

Story 2

As a staff member, I want to search for a patient by ID so that I can quickly locate their information.

Acceptance criteria:

- Given a patient record exists, when the staff member enters the correct patient name, then the system shall display the matching patient record.
- Given no patient has the entered name, when the staff member performs the search, then the system shall indicate that no matching patient was found.

Story 3

As a staff member, I want to book an appointment for a patient with a practitioner so that appointments can be managed electronically.

Acceptance criteria:

- Given a patient and practitioner exist and the practitioner is available, when the staff member selects a valid date and time and confirms the booking, then the system shall create the appointment.
- Given the practitioner already has an appointment at the selected time, when the staff member attempts to create another appointment at that time, then the system shall reject the booking and indicate that the time is unavailable.

Story 4

As a staff member, I want to cancel an appointment so that the clinic can keep appointment information up to date.

Acceptance criteria:

- Given an appointment exists, when the staff member selects cancel, then the appointment status shall change to cancelled.
- Given an appointment has already been cancelled, when the staff member attempts to cancel it again, then the system shall not create a duplicate cancellation and shall retain the existing cancelled status.

Story 5

As a practitioner, I want to view my appointment schedule so that I can see my upcoming appointments and availability.

Part F - AI Requirements Review: AI ON

Prompt: Act as a software requirements reviewer. Review the SmartCare requirements for ambiguity, inconsistency, missing clarification questions and testability. Do NOT invent new client requirements. For every suggestion, state whether it is based on evidence or is only a question/assumption requiring validation.

Requirement / Area	Issue identified	Evidence or validation?	Suggested action
Patient search	The requirements use both patient ID and patient name in different sections.	Evidence of inconsistency — your current requirements conflict with each other.	Choose one consistently, but whether the system should search by ID or name requires client validation because the case study does not specify the search method.
FR-01 – Create patient record	Generally clear, but “patient record” could contain different information.	Question requiring validation	Confirm what information must be stored in a patient record.
FR-02 – Search patient	“Search by name”/“search by ID” must be consistent.	Evidence + validation	Make the requirement and acceptance criteria use the same search method.
FR-05 – Prevent duplicate bookings	Clear about preventing two appointments for the same practitioner at the same time.	Evidence — duplicate bookings are explicitly identified as a SmartCare problem.	Keep the requirement.
FR-06 – Practitioner availability	“Availability” could mean available time slots, working hours, or something else.	Question requiring validation	Confirm what information about availability must be displayed.
FR-09 – Basic operational reports	“Basic” is ambiguous because the case study does not define which reports are required.	Question requiring validation	Ask the client which operational information/reports are required. Do not invent specific reports.
NFR-01 – Reliability	“Without crashing” is difficult to test and does not directly specify what acceptable reliability means.	Evidence of ambiguity	Focus the wording on maintaining data without unexpected loss during normal operation.
NFR-02 – Usability	“Simple task” and “without	Evidence of	Define what successful usability

Requirement / Area	Issue identified	Evidence or validation?	Suggested action
	technical knowledge” are subjective and difficult to measure.	ambiguity	means through client-approved criteria.
NFR-03 – Data integrity	This is reasonably testable because appointments cannot reference non-existent patients/practitioners.	Evidence — preventing inconsistent appointment information supports the case-study problems.	Keep, with clearer wording.
NFR-04 – Maintainability	Modular design is clear but does not define what level of modularity is expected.	Question requiring validation	Confirm whether the stated modularity is sufficient for the client's maintenance expectations.
NFR-05 – Testability	“Core business logic” could be interpreted differently, but the intent is reasonably clear.	Evidence + minor ambiguity	Clarify what components are considered core business logic if necessary.
Cancelled appointments	Some sections treat retaining cancelled appointments as provisional.	Evidence of inconsistency — reliable appointment history is explicitly identified as a problem.	Treat retaining cancelled appointments as supported by the case study rather than an invented feature.
Staff/receptionist role	Requirements refer to “staff” and “receptionist” interchangeably.	Question requiring validation	Confirm which staff roles will perform each function.
Patient self-service	It is unclear whether patients themselves can book/cancel appointments.	Question requiring validation — the case study does not explicitly state this.	Confirm whether booking/cancellation is staff-only or also available to patients.
Appointment cancellation	The requirement says an appointment can be cancelled, but does not specify who can cancel it or under what conditions.	Question requiring validation	Confirm who is authorised to cancel appointments and any applicable rules.
Appointment status	“Update the status” does not specify which statuses are	Question requiring validation	Confirm the allowed appointment statuses.

Requirement / Area	Issue identified	Evidence or validation?	Suggested action
	permitted.		

Part G - VERIFY the AI Review

Classify each significant AI suggestion as Accepted, Modified, Rejected, or Unverified. Explain the evidence used.

AI Suggestion	Classification	Evidence Used
Make patient search method consistent	Modified	The review identifies a real inconsistency: some sections use patient ID while others use patient name.
Define what information a patient record contains	Unverified	FR-01 is clear about how to create a record, but the detail is unknown.
Make FR-02 and its acceptance criteria use the same search method	Modified	The review found that the requirement and acceptance criteria currently conflict.
Keep FR-05 duplicate booking prevention	Accepted	Duplicate bookings are identified as a SmartCare problem and this fixes it.
Clarify practitioner availability	Unverified	“Availability” could mean different things and the review states that the exact information to display requires validation.
Clarify basic operational reports	Unverified	Reports are required but aren’t specified which reports.
Change NFR-01 from “without crashing” to focus on data preservation	Modified	The review identifies “without crashing” as ambiguous and difficult to test. The wording should be made more measurable.
Clarify NFR-02 usability	Modified	“Simple task” and “without technical knowledge” are subjective, so the requirement needs clearer testable criteria.
Keep NFR-03 data integrity	Accepted	The requirement is reasonably testable and addresses consistency between patients, practitioners and appointments.
Clarify NFR-04 modularity	Unverified	The required level of modularity is not specified and needs client validation.
Clarify NFR-05 core business logic	Unverified	“Core business logic” could be interpreted differently, so

		clarification may be required.
Treat retaining cancelled appointments as supported	Accepted	The case study identifies a lack of reliable appointment history as a problem, supporting retention of cancelled appointments.
Clarify the staff/receptionist role	Unverified	The requirements use both terms, but the exact staff roles responsible for functions are not established.
Clarify patient self-service	Unverified	The requirements do not establish whether patients can book or cancel their own appointments.
Clarify who can cancel appointments	Unverified	FR-07 says staff can cancel an appointment but does not establish specific authorisation or cancellation conditions.
Define permitted appointment statuses	Unverified	FR-08 allows status updates but does not specify which statuses are permitted.

Part H - Finalise SmartCare v0.2

Submit stakeholder analysis, scope, 8-12 FRs, 4-6 NFRs, 4-6 user stories, acceptance criteria, assumptions/open questions and selected AI review evidence.

Reflection

In 150-250 words: What did AI notice that you missed? What did AI invent or overreach on? Which requirement changed after review? Why must requirements have evidence?

AI identified several issues that I had initially overlooked when developing the SmartCare requirements. The main issue was ambiguity in terms such as “basic operational reports”, “practitioner availability”, “without crashing” and “simple task”, which could make the requirements difficult to test consistently. It also highlighted that the allowed appointment statuses and staff responsibilities had not been clearly defined.

AI also overreached by suggesting features such as SMS reminders, online payments and AI treatment recommendations. These features were not supported by the SmartCare case study or the functional requirements and would unnecessarily expand the scope of the initial system. Other suggestions, such as a specific receptionist role and patient self-service, could be reasonable but were unverified and therefore require client validation.

Following the review, the patient search requirement was changed so that the search method is consistent across the functional requirement, user story and acceptance criteria. NFR-01 was also improved to focus on maintaining data without unexpected loss rather than simply “not crashing”.

Requirements must have evidence because they should represent verified client needs rather than assumptions. Evidence ensures that each requirement can be justified, tested and traced back to the case study or stakeholder needs.

